

Total No. of Questions : 8]

SEAT No. :

**PE2746**

[Total No. of Pages : 2

[6583] 328

**T.E. (Robotics) (Honors)**

**304181 HR : PRINCIPLES OF ROBOTICS - I**

**(2019 Pattern) (Semester - V)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Draw sketches wherever required.
- 3) Figures to the right indicate full marks.

- Q1)** a) Explain design considerations of gripper selection. [5]  
b) Classify grippers on various parameters. [6]  
c) Explain mechanical grippers with neat sketch. [6]

OR

- Q2)** a) Explain vacuum grippers with neat sketch. [5]  
b) Explain various tools used as end effectors. [6]  
c) Explain magnetic grippers with sketch. [6]

- Q3)** a) Define a Robotic sensor. State desirable features of sensors. [6]  
b) Explain range sensors with neat sketch. [6]  
c) What is compliance used in Robotics? What is significance of compliance? [6]

OR

- Q4)** a) Explain important characteristics of Robot Sensors. [6]  
b) Explain capacitive sensor with neat sketch. [6]  
c) Explain with neat sketch LVDT. [6]

*P.T.O.*

- Q5)** a) With the help of suitable diagram explain the significance of D-H- Parameter. [5]
- b) Write short Note on
- i) Singularity
  - ii) Kinematic Calibration [6]
- c) Write short note on Jacobian transformation with one example. [6]

OR

- Q6)** a) A 2 DOF planar R-R Manipulator has  $L_1 = 120$  mm &  $L_2 = 75$ mm. Determine joint angles using geometric approach, so that face end is located at (100, 70). [6]
- b) With the help of suitable illustration, explain difference between forward & Inverse kinematics. [5]
- c) Explain with neat sketch D-H Parameters. [6]
- Q7)** a) Explain economical aspects of Robot Design. [6]
- b) Write a short Note of Robot Safety. [6]
- c) Explain 3 major functions of Machine Vision systems alongwith diagram. [6]

OR

- Q8)** a) Explain application of robotics is areas of [6]
- i) Health care
  - ii) Research & exploration
  - iii) Various industry
  - iv) Home sector
- b) Explain Robot vision systems. Also explain 3 levels of processing. [6]
- c) Explain various Image processing techniques. [6]

